Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

LEARNER GUIDE

For

Data Analytics and Visualization with Power BI

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Conducted by

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
14.0	01 July 2024	Legacy Power BI courseware and exercise set.	Tertiary Infotech
15.0	20 August 2026	Rebuilt visual deck; aligned detailed Learner Guide, slide-mapped Lesson Plan, ten self-contained labs, editable PowerPoint charts, current WSQ administration and assessment mapping.	Dr Alfred Ang

TABLE OF CONTENTS

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How to Use This Guide

This guide explains the Power BI concepts used in the visual slide deck and provides the detailed click-by-click procedures deliberately kept out of the PowerPoint. Work through the labs in order: each reuses an approved starter PBIX, a lab-specific Excel intake/control workbook, and an evidence checklist.

- Windows laptop with current Power BI Desktop
- Access to Power BI Service for dashboard tasks
- Complete lab folders downloaded locally
- Permission to use only the supplied training data

Learning Outcomes and WSQ Mapping

Outcome	Learner capability	TSC mapping
LO1	Perform data transformation techniques using Power BI	K1 · A1 · LO1
LO2	Create data visualizations that portray patterns, trends and correlations	K2 · A2 · LO2
LO3	Develop dashboards and scorecards for decision monitoring	K3 · A3 · LO3
LO4	Create dynamic and interactive data visualizations	K4 · A4 · LO4
LO5	Create a data model for strategic visualization and mapping	K5 · A5 · LO5
LO6	Perform data analytics and queries with DAX	A6 · LO6

Topic 1: Get Started with Power BI and Power Query

Deck reference: Slides 17–38 · K1 · A1 · LO1

1.1 From data to action

Descriptive, diagnostic, predictive, prescriptive and cognitive analytics support different decisions.

1.2 Roles in the data team

Business experts frame the question; analysts transform evidence; engineers govern dependable data flows.

1.3 The analyst workflow

Prepare → model → visualize → analyze → publish → monitor; each stage leaves evidence for review.

1.4 Power BI ecosystem

Desktop authors reports and semantic models; Service distributes, refreshes, governs and collaborates.

1.5 Dataset, semantic model, report, dashboard

Separate reusable data logic from multi-page analysis and single-page monitoring.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

1.6 Desktop views

Report view composes visuals, Data view inspects loaded tables, and Model view manages relationships.

1.7 Connection choices

Files, folders, databases, web sources and cloud services differ in refresh, scale and governance.

1.8 Import or DirectQuery

Import favors speed and modeling freedom; DirectQuery favors source currency but adds dependency and latency.

1.9 Navigator decisions

Select only required tables and preview structure before loading to reduce noise and model size.

1.10 Power Query Editor

Applied Steps make transformations visible, ordered, repeatable and auditable.

1.11 Promote headers and remove rows

Fix structural metadata before analytical logic so column names and data boundaries are reliable.

1.12 Choose and rename columns

Keep only decision-relevant fields and give them business-friendly names.

1.13 Pivot and unpivot

Wide layouts are human-friendly; long tidy layouts are usually better for filtering, aggregation and visuals.

1.14 Replace values, nulls and errors

Differentiate genuine zero, missing, not applicable and invalid data before replacement.

1.15 Remove duplicates

Define the business key before deduplication; repeated rows are not always duplicates.

1.16 Data types and locale

Types determine aggregation and date behavior; locale determines how text values are interpreted.

1.17 Append queries

Stack tables with the same grain and schema, then reconcile before/after row counts.

1.18 Merge queries

Join tables by validated keys; left outer, inner and full outer joins answer different integrity questions.

1.19 Column profiling

Quality, distribution and profile views surface nulls, errors, outliers and unexpected cardinality.

1.20 M and Advanced Editor

Power Query records transformations as M; readable steps improve reuse and troubleshooting.

Lab 1: Connect, Inspect and Profile an Excel Dataset · Slide 37

Goal: A profiled query with correct headers, types and documented quality issues

Files: labs/01-data-intake/Retail_Intake.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix` in Power BI Desktop and save a working copy as `Lab01-<your-name>.pbix`.
2. Select **Home > Get data > Excel workbook** and choose `Retail_Intake.xlsx`.
3. In Navigator, select `Orders`, `Returns`, and `People`; review the preview before selecting **Transform Data**.
4. In Power Query Editor, confirm that the first row is used as headers and that each query has a meaningful name.
5. Turn on **View > Column quality**, **Column distribution**, and **Column profile**.
6. Record null, error, and distinct-value observations in the workbook `Evidence` sheet.
7. Set date, whole-number, decimal, and text data types deliberately; use locale where required.
8. Remove columns that do not support the stated retail questions, but keep all business keys.
9. Rename technical columns only when the new name remains unambiguous and consistent.
10. Select **Close & Apply**, then confirm all three tables appear in Data view.
11. Create a card for total Sales and a table showing Order ID, Customer Name, Sales, and Profit.
12. Save the PBIX and capture the query list, column-profile evidence, and the first report page.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- All required tables load; column quality is reviewed; no unintended errors remain.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Lab 2: Clean, Transform, Append and Merge · Slide 38

Goal: A repeatable Power Query transformation pipeline with an audit trail

Files: labs/02-power-query/Retail_Quality_Issues.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy, and import `Retail_Quality_Issues.xlsx` through **Transform Data**.
2. Rename the imported query `Sales_Staging` and inspect Applied Steps before adding transformations.
3. Remove the two note rows above the actual header and promote the remaining first row as headers.
4. Trim and clean text columns; standardize region values so `South East`, `SE`, and `Southeast` become `Southeast`.
5. Replace blank discount values only after deciding whether blank means zero or unknown; document the choice.
6. Set Order Date using the correct locale and isolate conversion errors rather than deleting them silently.
7. Define the business key, identify duplicate records, and keep the intended surviving record.
8. Unpivot the monthly target columns into `Month` and `Target`.
9. Append the `Late_Arrivals` query to the cleaned sales query and reconcile row counts.
10. Merge the `Region_Manager` table with a left outer join; expand only Manager and Territory.
11. Create a conditional column for Margin Band using the thresholds in the workbook `Control` sheet.
12. Review the generated M in Advanced Editor and add a short comment above the business-rule step.
13. Close & Apply, then validate the Control totals and complete the Evidence sheet.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- Headers, types, nulls and duplicates are corrected; append and merge row counts reconcile.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Topic 2: Create Data Visualizations and Reports

Deck reference: Slides 39–60 · K2 · A2 · LO2

2.1 Start with the question

Choose a visual only after naming the comparison, trend, distribution, relationship or composition to reveal.

2.2 Column and bar charts

Use length on a common baseline for accurate category comparison; sort to make ranking visible.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

2.3 Line and area charts

Use lines for change over ordered time; use area sparingly when magnitude and composition both matter.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

2.4 Cards, KPIs and gauges

Cards state a value, KPIs add direction and targets, and gauges show progress toward one bounded goal.

2.5 Combo charts

Combine columns and a line when two measures share a meaningful category and their scales remain interpretable.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

2.6 Decomposition tree

Let users drill across candidate dimensions to explain a result without predefining one path.

2.7 Pie, doughnut and funnel

Use part-to-whole displays for few categories and funnels only for genuinely ordered stages.

2.8 Maps and choropleths

Use symbols for magnitude and filled regions for rates; normalize before comparing unequal geographies.

2.9 Tables and matrices

Tables preserve precise records; matrices add hierarchy, subtotals and cross-tab analysis.

2.10 Scatter and bubble charts

Position reveals correlation, while size and color add variables; association is not causation.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

2.11 Ribbon, treemap and waterfall

Show rank change, hierarchical composition and additive bridges when those structures answer the question.

2.12 Q&A and smart narratives

Natural-language and generated summaries accelerate exploration but still require human validation.

2.13 Numeric, grid and performance visuals

Match information density to the decision: glanceable status, precise lookup or target tracking.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

2.14 Change and resize visuals

Changing type can change meaning; protect labels, proportions and hierarchy when reflowing a page.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

2.15 Format for comprehension

Use consistent color roles, restrained emphasis and direct labels to reduce decoding effort.

2.16 Titles and backgrounds

Insight titles state the takeaway; backgrounds should organize, not decorate.

2.17 Tooltips

Use tooltips for supporting context, definitions and secondary measures without crowding the main canvas.

2.18 Drill down and expand

Hierarchies support movement between levels; keep the current level and path visible.

2.19 Themes

A theme standardizes color, type and defaults so reports remain coherent and maintainable.

2.20 Accessibility

Reading order, alt text, contrast, keyboard focus and non-color cues make insight available to more users.

Lab 3: Choose Visuals for Trends and Correlations • Slide 59

Goal: A one-page analytical report using fit-for-purpose native visuals

Files: labs/03-visual-choice/Sales_Trends.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy, and import `Sales_Trends.xlsx`.
2. Read the five stakeholder questions on the `Questions` sheet before adding any visual.
3. Create a clustered bar chart for ranked category comparison and sort by Sales descending.
4. Create a line chart for monthly Sales and use a continuous date axis.
5. Create a scatter chart with Sales on X, Profit on Y, Quantity as size, and Category as legend.
6. Create one KPI or gauge that compares current Sales with Target Sales.
7. Use a matrix where precise product and region values are required.
8. For each visual, replace the generic title with a short insight title that answers its question.
9. Use color consistently: one accent for focus, neutrals for context, and red only for a defined exception.
10. Check axes for misleading truncation and confirm currency/percentage formats.
11. Add a small text box stating one limitation of the comparison or correlation shown.
12. Complete the visual-choice rationale and evidence checklist in the workbook.

Evidence and acceptance

- Saved working PBIX with learner name

- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- Each visual answers a stated question; encodings are accurate; titles express insight.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Lab 4: Design an Accessible Report Story · Slide 60

Goal: A polished report page with hierarchy, theme, tooltips and accessible navigation

Files: labs/04-report-story/Customer_Experience.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy, and import `Customer_Experience.xlsx`.
2. Apply the provided `Theme` color roles consistently across the page.
3. Create a page title that states the decision and a subtitle that states the active period.
4. Arrange KPI cards first, trend evidence second, and diagnostic detail last in the reading order.
5. Create a tooltip page showing response volume, satisfaction score, and verbatim-comment count.
6. Add a drill hierarchy from Region to Store and test drill down, expand, and drill up.
7. Add alt text to every meaningful visual and mark decorative elements as decorative where supported.
8. Open **View > Selection** and **Tab order**; place objects in a logical keyboard reading sequence.
9. Check text and data colors for sufficient contrast and ensure status is never shown by color alone.
10. Use whitespace and alignment instead of borders around every object.
11. Test the page at 100% and Fit to page, then capture a screenshot for Evidence.
12. Ask a peer to state the main finding in ten seconds; revise if their answer differs from the intended message.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- The page has a clear reading order, consistent formatting, alt text and no misleading scale.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Topic 3: Develop Dashboards and Scorecards

Deck reference: Slides 61–78 · K3 · A3 · LO3

3.1 Report versus dashboard

Reports are multi-page interactive analysis; dashboards are one-page monitoring surfaces in Power BI Service.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

3.2 Dashboard development cycle

Define decisions → select KPIs → design tiles → publish → test → monitor and refine.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

3.3 Pin a visual

Pinned tiles create a stable monitoring surface while preserving a path back to the source report.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

3.4 Pin a live page

A live page retains report interactions; use it when monitoring needs more than independent tiles.

3.5 Combine multiple reports

A dashboard can assemble governed signals from several reports into one executive view.

3.6 Alerts

Alerts turn threshold breaches into action; they require clear owners, frequency and escalation rules.

3.7 Ask questions

Dashboard Q&A supports fast exploration when model names, synonyms and relationships are business-friendly.

3.8 Quick insights

Automated patterns are hypotheses to review, not conclusions to accept without context.

3.9 Dashboard themes

Consistent themes support recognition across tiles without overwhelming status colors.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

3.10 Mobile layout

Prioritize the few signals and actions needed on a small screen; do not merely shrink desktop design.

3.11 Scorecards and goals

A scorecard adds targets, owners, status and check-ins to make performance accountable.

3.12 Workspaces

Workspaces separate authoring, review and distribution responsibilities.

3.13 Apps and sharing

Apps package approved content for audiences; direct sharing is narrower and harder to govern at scale.

3.14 Refresh and gateways

Refresh design connects source availability, credentials, schedules and failure monitoring.

3.15 Dashboard storytelling

Arrange tiles from outcome to drivers to exceptions so executives can scan, diagnose and act.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

3.16 Dashboard QA

Verify current data, tile links, mobile view, ownership, permissions and decision usefulness before release.

Decision check: state the stakeholder question, the metric and grain, the active filter context, the visible finding, and the action the display should support.

Lab 5: Build an Executive Dashboard and Scorecard · Slide 78

Goal: A Sales Monitoring dashboard and KPI scorecard in Power BI Service

Files: labs/05-dashboard-scorecard/Executive_KPIs.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy, and import `Executive_KPIs.xlsx`.
2. Create or verify measures for Sales, Profit, Margin %, Target Sales, and Target Attainment %.
3. Build a report page with one executive outcome row and one driver/exceptions row.
4. Publish the working report to the training workspace in Power BI Service.
5. Create a dashboard named `Sales Monitoring - <your initials>`.
6. Pin the Sales, Profit and Margin visuals as individual tiles.
7. Pin the decision page as a live page only if its interactions are needed for monitoring.
8. Resize and arrange tiles from outcome to drivers to exceptions.
9. Create a scorecard with at least two goals, targets, owners and status logic.
10. Configure an alert on one supported KPI tile and document the owner and response action.
11. Review the mobile layout and move only the essential monitoring signals above the fold.
12. Capture the report, dashboard, scorecard and mobile layout as Evidence; do not expose confidential workspace data.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- Tiles and targets are current, decision-focused and traceable to the report.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Topic 4: Create Dynamic and Interactive Visualizations

Deck reference: Slides 80–97 · K4 · A4 · LO4

4.1 Interaction model

Selections propagate through relationships and visual interactions; every response should be intentional.

Model check: confirm one row's meaning, the key uniqueness rule, relationship cardinality, filter direction and how totals should behave before building a visual.

4.2 Filter scope

Dataset security, report filters, page filters, visual filters and drillthrough each control a different boundary.

4.3 Cross-filter

A selection removes unrelated marks so the page focuses on the chosen segment.

4.4 Cross-highlight

A selection highlights contribution while retaining the whole for comparison.

4.5 Edit interactions

Configure each visual to filter, highlight or ignore the source selection based on the stakeholder question.

4.6 Slicers

Expose the few high-value filters users need and show the active selection clearly.

4.7 Sync slicers

Shared filter state creates continuity across pages; avoid hidden states that surprise users.

4.8 Hierarchies

Ordered levels support drill paths such as Year → Quarter → Month or Region → Country → Store.

4.9 Drillthrough

Carry context from a summary mark to a focused detail page with a clear route back.

4.10 Bookmarks

Capture filter, display and navigation states to build guided stories and reset controls.

4.11 Buttons and navigation

Use consistent labels, states and placement so navigation feels predictable.

4.12 Conditional formatting

Use rules to direct attention to exceptions, not to decorate every value.

4.13 Report-page tooltips

Provide rich context on hover while keeping the primary view simple.

4.14 Field parameters

Let users switch dimensions or measures without duplicating pages.

4.15 What-if parameters

Turn a report into a controlled scenario tool while distinguishing assumptions from actuals.

4.16 Interactive UX

Preserve orientation with page titles, breadcrumbs, reset state and visible filter context.

4.17 Stakeholder review

Test the display against real questions, expected actions and common misinterpretations.

Lab 6: Create Dynamic Interactions and Navigation · Slide 97

Goal: An interactive report with slicers, drillthrough, bookmarks and tested interactions

Files: labs/06-dynamic-interactions/Campaign_Performance.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy, and import `Campaign_Performance.xlsx`.
2. Create a summary page with Channel, Campaign, Spend, Leads and Revenue visuals.
3. Add a date slicer and a Channel slicer; show the current selection clearly.
4. Configure one source visual to cross-filter a target and another to cross-highlight a comparison.
5. Disable an interaction where responding would mislead or create visual noise.
6. Create a hierarchy for Channel > Campaign > Creative and test all drill states.
7. Create a drillthrough page for Campaign detail and add a Back button.
8. Create a report-page tooltip with conversion rate, cost per lead and return on ad spend.
9. Create bookmarks for Executive view, Analyst view and Reset filters.
10. Add buttons for the bookmarks and use consistent selected/hover states.
11. Add a field parameter that switches the main comparison between Revenue, Leads and Spend.
12. Run the interaction test matrix in the workbook and fix every unexpected state.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- Every interaction is intentional; drill paths work; reset navigation restores the default state.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Topic 5: Create a Strategic Data Model

Deck reference: Slides 98–116 · K5 · A5 · LO5

5.1 Semantic model

A model organizes reusable business meaning: tables, relationships, measures, hierarchies and security.

Model check: confirm one row's meaning, the key uniqueness rule, relationship cardinality, filter direction and how totals should behave before building a visual.

5.2 Star schema

A central fact table records events; surrounding dimensions describe who, what, where and when.

Model check: confirm one row's meaning, the key uniqueness rule, relationship cardinality, filter direction and how totals should behave before building a visual.

5.3 Fact tables

Facts have a declared grain, foreign keys and numeric observations suitable for aggregation.

5.4 Dimension tables

Dimensions provide stable descriptive attributes for grouping, filtering and labeling.

5.5 Relationship keys

Use unique dimension keys and matching fact foreign keys; validate unmatched and duplicate values.

Model check: confirm one row's meaning, the key uniqueness rule, relationship cardinality, filter direction and how totals should behave before building a visual.

5.6 Cardinality

One-to-many is the normal star pattern; one-to-one and many-to-many require explicit justification.

5.7 Cross-filter direction

Single direction is easier to reason about; bidirectional filters can create ambiguity and unexpected totals.

5.8 Many-to-many

Bridge tables make shared membership explicit and prevent uncontrolled duplication.

5.9 Date dimension

A contiguous date table enables consistent calendar attributes and time intelligence.

5.10 Table and column properties

Names, formats, data categories and default summarization shape the authoring experience.

5.11 Hierarchies

Curated hierarchies guide valid drill paths and reduce repeated setup.

5.12 Measures

Measures calculate at query time and respond to filter context.

Calculation check: test the result under at least two filters, reconcile it to the Excel Control sheet, and document how blanks and zero denominators are handled.

5.13 Calculated columns

Columns calculate row by row at refresh and increase model size; use them only when a stored attribute is needed.

Calculation check: test the result under at least two filters, reconcile it to the Excel Control sheet, and document how blanks and zero denominators are handled.

5.14 Granularity

Declare one row's meaning before combining sources or writing measures.

5.15 Role-playing dimensions

One Date table can relate to order, ship and due dates; only one relationship is active at a time.

5.16 Model performance

Remove unused columns, prefer star schemas, reduce high cardinality and keep transformations upstream where practical.

Model check: confirm one row's meaning, the key uniqueness rule, relationship cardinality, filter direction and how totals should behave before building a visual.

5.17 Business-friendly model

Hide technical keys, group measures and use descriptions so authors find the right fields.

Model check: confirm one row's meaning, the key uniqueness rule, relationship cardinality, filter direction and how totals should behave before building a visual.

5.18 Model QA

Check uniqueness, referential integrity, cardinality, filter paths, totals and refresh behavior before visualization.

Model check: confirm one row's meaning, the key uniqueness rule, relationship cardinality, filter direction and how totals should behave before building a visual.

Lab 7: Model a Star Schema and Date Dimension · Slide 116

Goal: A semantic model with fact/dimension relationships and a marked date table

Files: labs/07-star-schema/Star_Schema.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy, and import every table from `Star_Schema.xlsx`.

2. In Power Query, confirm each table grain and ensure dimension keys are unique.
3. In Model view, place `FactSales` in the center and the Date, Product, Customer and Region dimensions around it.
4. Create one-to-many relationships from each dimension key to its matching fact foreign key.
5. Use single cross-filter direction unless the scenario documents a tested exception.
6. Mark `DimDate` as the date table using its Date column.
7. Create a Month Sort column and sort Month Name by Month Number.
8. Create a Region > Country > Store hierarchy and a Product Category > Product hierarchy.
9. Hide surrogate keys and other technical columns from Report view.
10. Set currency, percentage, date, and default summarization properties.
11. Use the Orphans sheet to validate that every fact foreign key finds one dimension row.
12. Build a small matrix and slicer test page to prove totals and filter paths are correct.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- Relationships use correct keys/cardinality; filters flow predictably; model ambiguity is zero.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Topic 6: Analyze Data with DAX

Deck reference: Slides 117–141 · A6 · LO6

6.1 DAX purpose

DAX defines reusable calculations over a semantic model; it is not a replacement for data cleaning.

Calculation check: test the result under at least two filters, reconcile it to the Excel Control sheet, and document how blanks and zero denominators are handled.

6.2 Formula syntax

A calculation has a name, expression, functions, operators and references to tables, columns or measures.

6.3 Reference rules

Columns require table context; measures should be referenced by measure name to preserve reusable logic.

6.4 Data types and operators

Type compatibility controls comparison, arithmetic and blank handling.

6.5 Row context

Calculated columns and iterators evaluate one row at a time.

6.6 Filter context

Visual axes, slicers, filters and CALCULATE define the subset a measure evaluates.

6.7 CALCULATE

CALCULATE evaluates an expression under modified filter context and is central to analytical DAX.

Calculation check: test the result under at least two filters, reconcile it to the Excel Control sheet, and document how blanks and zero denominators are handled.

6.8 Variables

VAR and RETURN improve readability, avoid repeated evaluation and simplify debugging.

6.9 Implicit and explicit measures

Explicit measures centralize definitions, formatting and governance.

Calculation check: test the result under at least two filters, reconcile it to the Excel Control sheet, and document how blanks and zero denominators are handled.

6.10 Core measures

Start with additive Sales, Cost and Quantity, then derive Profit and Margin from governed measures.

Calculation check: test the result under at least two filters, reconcile it to the Excel Control sheet, and document how blanks and zero denominators are handled.

6.11 Safe division

DIVIDE handles zero or blank denominators more safely than the division operator.

6.12 Share of total

REMOVEFILTERS or ALL can create a denominator while retaining intentional dimensions.

6.13 Date table

A marked date table with continuous dates underpins consistent time calculations.

6.14 Time intelligence

YTD, prior period and growth measures depend on valid date relationships and filter context.

6.15 TOPN

TOPN returns a table ranked by an expression; pair it with clear ties and filter behavior.

6.16 Groups and bins

Grouping creates analytical segments; bins reveal distribution at a chosen resolution.

6.17 Outliers and anomalies

Flag exceptions using statistical context, then investigate causes rather than treating detection as explanation.

6.18 Clustering

Clusters suggest similar records based on selected variables; business review is needed to name and use them.

6.19 Analyze feature

Automated increase/decrease explanations provide candidate drivers that require validation.

6.20 What-if analysis

Disconnected parameter tables let users test assumptions without changing actual data.

6.21 Trend lines and forecasts

Trend and forecast overlays summarize direction and uncertainty; they do not prove causality.

6.22 Align findings with SMEs

Translate measures into a claim, supporting evidence, limitations and an action for domain review.

Lab 8: Create Core DAX Measures · Slide 139

Goal: Validated measures for Sales, Cost, Profit, Margin and share of total

Files: labs/08-dax-measures/Sales_Model.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy, and import `Sales_Model.xlsx`.
2. Create a display folder named `Core Measures` and keep all new measures in a dedicated Measures table.
3. Create `Total Sales = SUM(FactSales[Sales])`.
4. Create `Total Cost = SUM(FactSales[Cost])` and `Profit = [Total Sales] - [Total Cost]`.
5. Create `Profit Margin = DIVIDE([Profit], [Total Sales])` and format it as a percentage with one decimal place.
6. Create `Order Quantity = SUM(FactSales[Quantity])`.
7. Create `Sales % All Region = DIVIDE([Total Sales], CALCULATE([Total Sales], REMOVEFILTERS(DimRegion)))`.
8. Use variables to create an `Average Order Value` measure from sales and distinct order count.
9. Build a matrix by Region and Category and add the measures in a deliberate order.
10. Test each measure with and without slicers to confirm it responds to filter context.
11. Reconcile Total Sales, Profit and Margin to the workbook Control sheet.
12. Document one filter-context test and one division-by-zero test in Evidence.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- Measures reconcile to workbook controls and respond correctly to filter context.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Lab 9: Apply Time Intelligence and Scenario Analysis · Slide 140

Goal: YTD, prior-period, Top N and what-if analysis with decision commentary

Files: labs/09-advanced-analytics/Scenario_Analysis.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy, and import `Scenario_Analysis.xlsx`.
2. Confirm the active relationship from DimDate to FactSales and mark the date table.
3. Create Sales YTD, Sales Prior Year and Sales Growth % measures.
4. Create a Top N parameter and use TOPN logic or a Top N visual filter to rank products.
5. Create bins for order value and use a histogram to inspect distribution.
6. Add a trend line to a scatter chart and state why the relationship is not proof of causation.
7. Use the Analyze feature on one increase or decrease and record the suggested driver.
8. Create a what-if parameter for Discount Adjustment from 0% to 15% in 1% increments.
9. Create Scenario Revenue and Scenario Profit measures that use the selected parameter.
10. Build Actual versus Scenario comparison cards and a sensitivity line chart.
11. Identify one anomaly or outlier and test whether it persists after changing filters.
12. Write a stakeholder note with finding, evidence, limitation and recommended action.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- Date logic is valid; parameters change results; findings state business meaning and limitations.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Lab 10: Deliver the GlobalTech Sales Performance Cockpit · Slide 141

Goal: A stakeholder-ready PBIX report, dashboard evidence and QA checklist

Files: labs/10-capstone/GlobalTech_Capstone.xlsx and starter.pbix

Step-by-step

1. Open `starter.pbix`, save a working copy as `GlobalTech-Capstone-<your-name>.pbix`, and import `GlobalTech_Capstone.xlsx`.
2. Read the stakeholder brief and convert each question into a required visual or interaction.
3. Transform the source tables and document at least four meaningful Applied Steps.
4. Build a star schema with validated one-to-many relationships and a marked date table.
5. Create governed measures for Sales, Profit, Profit Margin, Sales % All Region, Target Attainment and YTD Growth.
6. Create an Executive Overview page with outcome KPIs, a country/category comparison and an exception list.
7. Create a Regional Analysis page with a map or region comparison, matrix and drillthrough path.
8. Create a Product Analysis page with trend, Top N and contribution evidence.
9. Add slicers, reset navigation, tooltips and intentional cross-filter/highlight interactions.
10. Publish to the training workspace and create the `Sales Monitoring` dashboard with report links intact.
11. Complete the workbook Acceptance and Evidence sheets, including one limitation and one recommended action.
12. Run the final QA: refresh, relationships, totals, visual accuracy, accessibility, interactions, mobile layout and privacy.

Evidence and acceptance

- Saved working PBIX with learner name
- Readable query/model/report/dashboard evidence
- Completed Excel Evidence sheet
- Finding, evidence, limitation and recommended action
- The solution meets all six outcomes, answers stakeholder questions and passes the evidence checklist.

Troubleshooting: if the PBIX cannot find a source, use Data source settings > Change Source and point it to the Excel workbook in the same lab folder. If a result is blank or wrong, check data type, grain, relationship path and filter context before rewriting the measure.

Quick Reference

Task	Power BI location / pattern
Connect to Excel	Home > Get data > Excel workbook
Open Power Query	Home > Transform data
Profile columns	Power Query > View > Column quality / distribution / profile
Manage relationships	Model view > Manage relationships
Create measure	Modeling > New measure
Publish	Home > Publish

Task	Power BI location / pattern
Create dashboard	Power BI Service > Pin visual / New dashboard
Reset filters	Bookmark + Button, or reset in the filter pane

Assessment and Support

Assessment sequence: TRAQOM QR code on the LMS → Assessment Digital Attendance → Written Assessment (60 minutes) → Practical Performance (90 minutes) → Submit completed answers on the LMS → Sign the Assessment Summary Record.

Courseware and assessment: https://lms-tms.tertiaryinfotech.com/

Support: enquiry@tertiaryinfotech.com · +65 6100 0613 · www.tertiarycourses.com.sg